

ENGINEERING CANDIDATE TECHNICAL TESTS BACKEND

JULY 2025



OVERVIEW

Build a **console application** (preferably in **C#**) that simulates a basic **Lending Platform**. This task is designed to evaluate your problem-solving approach, use of clean code practices, and ability to model domain logic, as well as how effectively you work with AI tools.

Use AI assistance (ChatGPT, Claude, Copilot, etc.) as part of your process. As part of your submission, include a brief AI log, the key prompts you used and any notable iterations or corrections. Copy/paste is fine.

The test is **timeboxed to 1 hour** and is not expected to be production-ready. Focus on demonstrating your approach, and feel free to include notes about any assumptions, improvements, or considerations you'd make for a production version.



REQUIREMENTS

Inputs (User)

- Loan amount (in GBP)
- Asset value the loan is secured against
- Applicant's credit score (1–999)

Outputs (System)

- Loan decision: Whether the applicant is successful or declined
- Total number of applicants (grouped by success status)
- Total value of loans written to date
- Mean average Loan to Value (LTV*) across all applications

* Loan to Value (LTV) is the loan amount expressed as a percentage of the asset's value it is secured against.



BUSINESS RULES

General Limits

- Decline if loan is < £100,000 or > £1.5 million

If loan is $\geq$ £1 million

- LTV must be 60% or less
- Credit score must be $\geq$ 950

If loan is < £1 million:

- If LTV < 60% - Credit score must be $\geq$ 750
- If LTV < 80% - Credit score must be $\geq$ 800
- If LTV < 90% - Credit score must be $\geq$ 900
- If LTV $\geq$ 90% - Decline



SUBMISSION

Evaluation Criteria

- Correctness of business logic
- Clarity and maintainability of code
- Separation of concerns and modularity
- Effective use of AI, quality of prompting, iteration, and critical review of AI output
- Ability to explain your reasoning, assumptions, and trade-offs

Submission Instructions

- Share a public Git repository containing your solution.
- Ensure all instructions for running the application are included.
- Include your AI log, key prompts, iterations, and any output you questioned or corrected



THANK YOU

